CIE AS PHYSICS (9702) • QUESTION PAPER

11.2 Fundamental Particles

Syllabus Section: 11. Particle Physics **Paper Type:** Structured Questions (Paper 2) **Total:** 7 Questions (53 Marks)

Question 1 Easy [10 marks]

Part (a) [3 marks]

A student is listing the flavours of quarks they can recall from memory. Their list is below:

- Bottom
- Strange
- Down

Name the flavours of quarks missing from their list.

Part (b) [4 marks]

Complete Table 1.1 below. The right hand column is in units of the elementary charge, *e*.

Table 1.1

Quark Flavour	Charge / *e*
Strange	
Charm	
Down	
Top	

Part (c) [2 marks]

Explain why a proton is not a fundamental particle.

Part (d) [1 mark]

Describe the difference between quarks and antiquarks.

Question 2 Easy [12 marks]

Part (a) [2 marks]

State the particles that form a meson.

Part (b) [3 marks]

A student proposes an undiscovered baryon with the following structure:

ttb

State whether it is possible for this baryon to exist.

Explain your answer.

Part (c) [3 marks]

Describe the changes to a proton's quark composition when it undergoes beta plus decay.

Part (d) [4 marks]

Show that an anti-proton has a relative charge of $-e$, where e is the elementary charge, using its quark composition.

Question 3 Medium [3 marks]

Part (a) [1 mark]

State the quark composition of a meson.

Part (b) [2 marks]

A hadron consists of two down quarks and a charm quark.

Determine the charge of the hadron. Show your working.

charge = ... C

Question 4 Medium [10 marks]

Part (a) [4 marks]

Carbon–10 $^{10}_{6}\mathrm{C}$ is an isotope that decays to an isotope of boron (symbol B) by the emission of a β^+ particle.

(i) Complete the nuclear equation for this decay, including the proton and nucleon numbers of all the particles involved.

$$^{10}_{6}\mathrm{C} \quad \rightarrow$$

[3]

(ii) During the decay, a quark in the carbon-10 nucleus changes into a different type.

State the change of quark that occurs.

[1]

Part (b) [2 marks]

State the two leptons in this decay.

Part (c) [3 marks]

State and explain why a hadron with charge –3*e* cannot exist.

e is the elementary charge.

Part (d) **[1 mark]**

State the possible quark combination for a hadron of charge +2*e*.

Question 5 Medium [8 marks]

Part (a) [2 marks]

State the quark composition of an alpha particle.

Part (b) [2 marks]

Use the quark model to show that the charge on the anti-neutron is 0.

Part (c) [3 marks]

A sigma baryon Σ is a hadron that consists of an up (u) quark, a down (d) quark and a strange (s) quark.

State the quark composition of Σ^+ , Σ^0 and Σ^- .

Part (d) [1 mark]

Explain why a baryon with the quark composition $u\bar{d}s$ cannot exist.

Question 6 Medium [7 marks]

Part (a) [1 mark]

State the number of 'up' quarks in the nucleus of $^{27}_{13}\mathrm{Al}$.

Part (b) [2 marks]

The D particle has a quark structure $u\bar{c}$.

Determine

(i) the classification of the D particle,

[1]

(ii) the charge on the D particle.

[1]

Part (c) [1 mark]

Using the quark model, show that the magnitude of the charge of a proton is equal to the charge of an electron.

Part (d) [3 marks]

Using the quark model for β^+ decay, show that charge is conserved in this decay.

Question 7 Medium [3 marks]

Part (a) [3 marks]

State the quark composition of:

(i) a proton

[1]

(ii) a neutron

[1]

(iii) an alpha-particle.

[1]